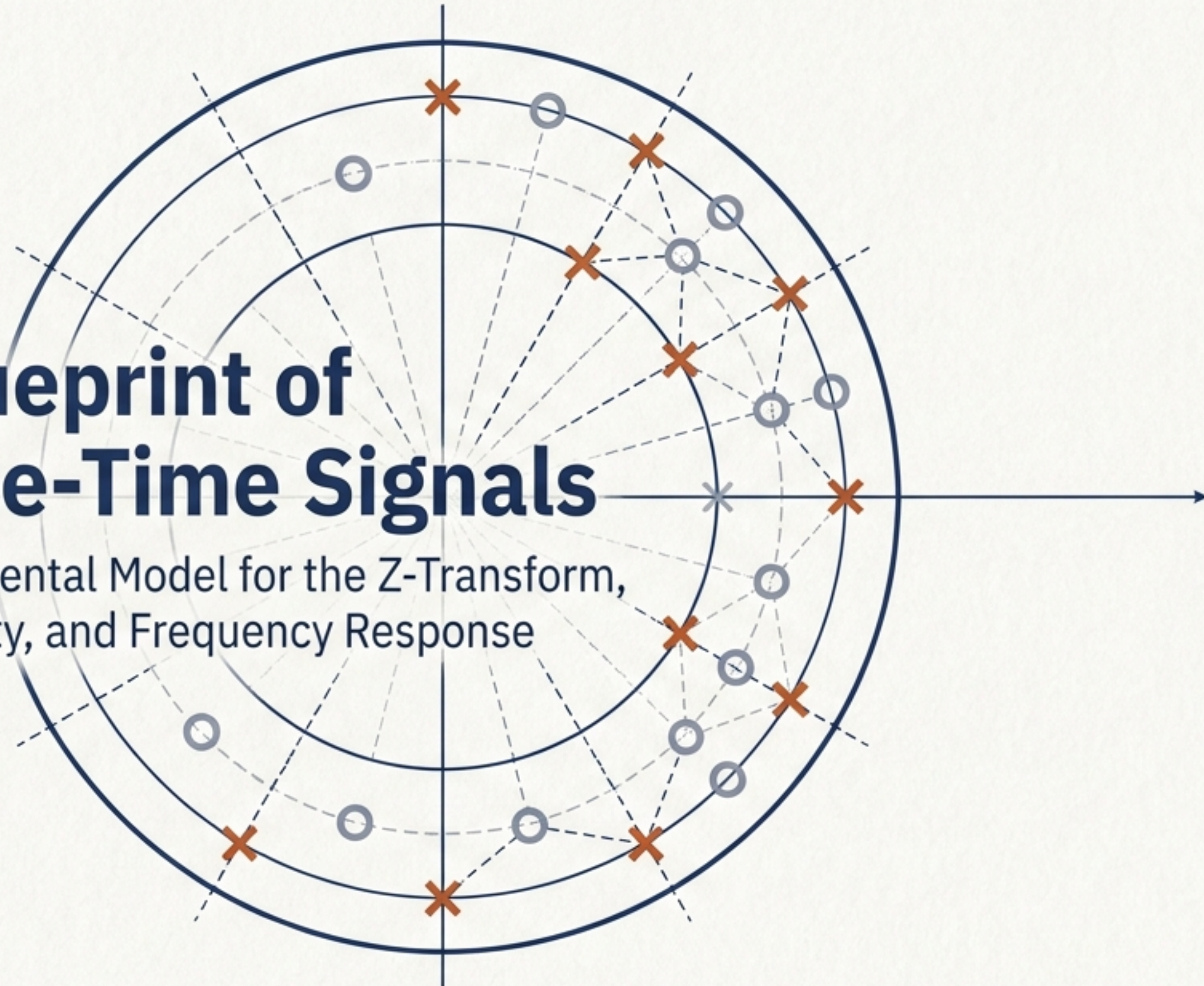


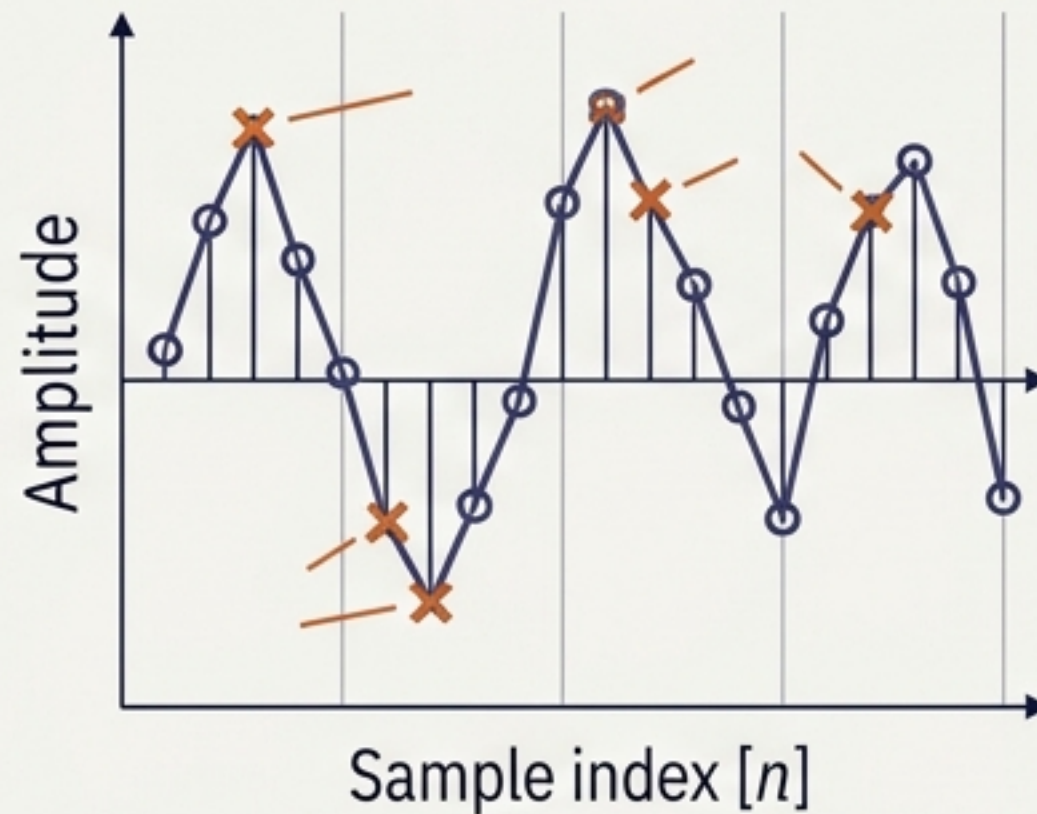
The Blueprint of Discrete-Time Signals

A Geometric Mental Model for the Z-Transform, System Stability, and Frequency Response



THE EDITORIAL DIAGNOSTIC BLUEPRINT: CLOSED-LOOP VISUAL LOGIC

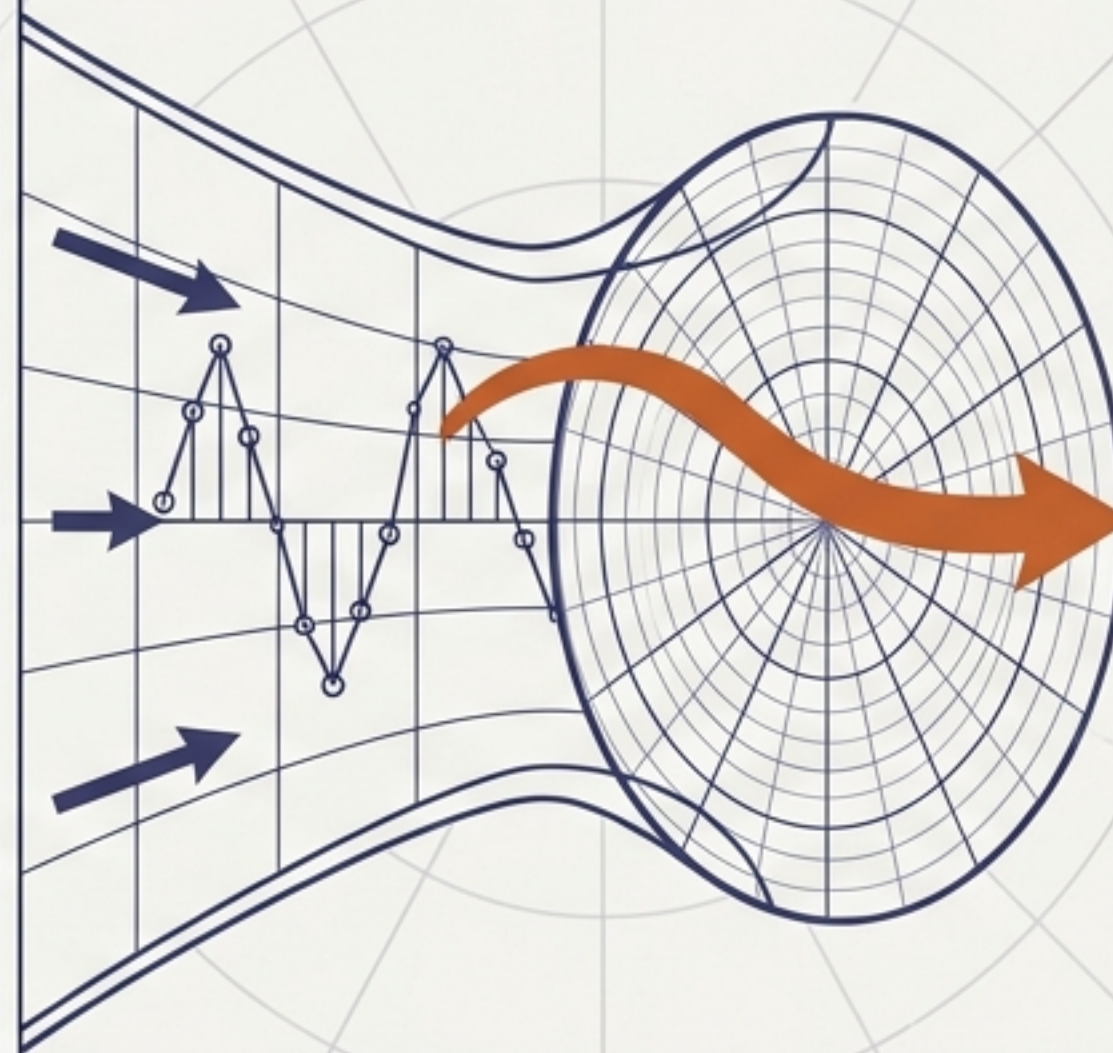
THE PROBLEM (TIME DOMAIN)



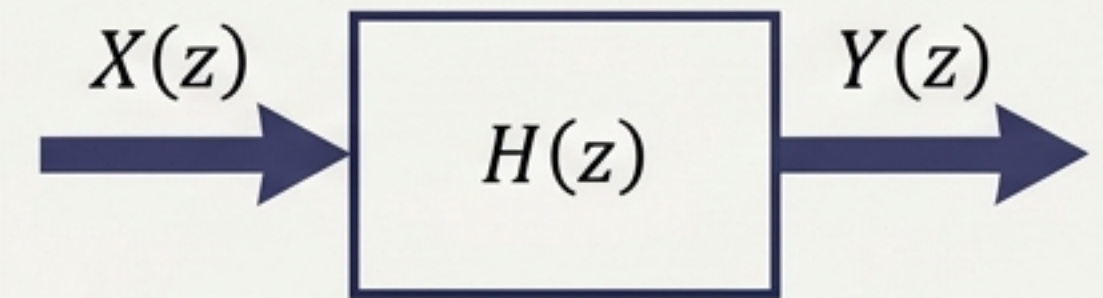
Difference equations are hard to solve directly.

$$y[n] - a y[n-1] = x[n]$$

THE PORTAL (Z)



THE SOLUTION (Z-DOMAIN)

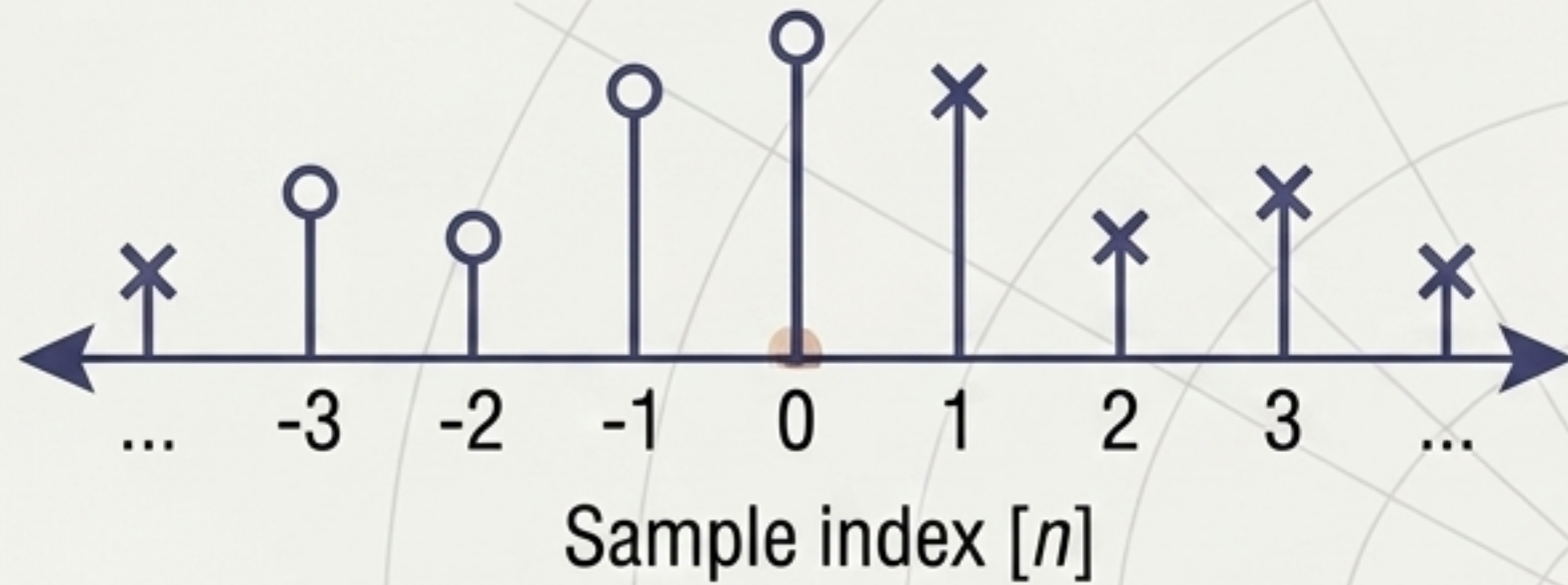


A difference equation becomes simple algebra. Delay in time becomes multiplication by z^{-1} .

$$Y(z)(1 - az^{-1}) = X(z) \rightarrow$$

$$H(z) = \frac{1}{1 - az^{-1}}$$

Bilateral Z-Transform



$$X(z) = \sum_{n=-\infty}^{+\infty} x[n]z^{-n}$$

Focus: General signal and system analysis.

Unilateral Z-Transform



$$X^+(z) = \sum_{n=0}^{+\infty} x[n]z^{-n}$$

Focus: Causal systems and initial conditions.

Insight: A delay adds an initial value term:

$$x[n-1] \leftrightarrow z^{-1}X^+(z) + x[-1]$$

$$z = r e^{j\omega}$$

Distance (r)

$r = |z|$ (Magnitude)

Rotation (ω)

$\omega = \arg(z)$ (Angle)

A constant magnitude $|z|=r$ forms a circle. The Unit Circle ($|z|=1$) is the most critical structure in discrete-time processing.

$X(z) + \text{ROC}$

The algebraic expression $X(z)$ is not enough. We must specify the Region of Convergence (ROC)—the set of all points where the infinite series converges to a finite value.

The Ambiguity Example

$$X(z) = \frac{1}{1 - az^{-1}}$$

If $|z| > |a|$

Represents a right-sided sequence:

$$x[n] = a^n u[n]$$

If $|z| < |a|$

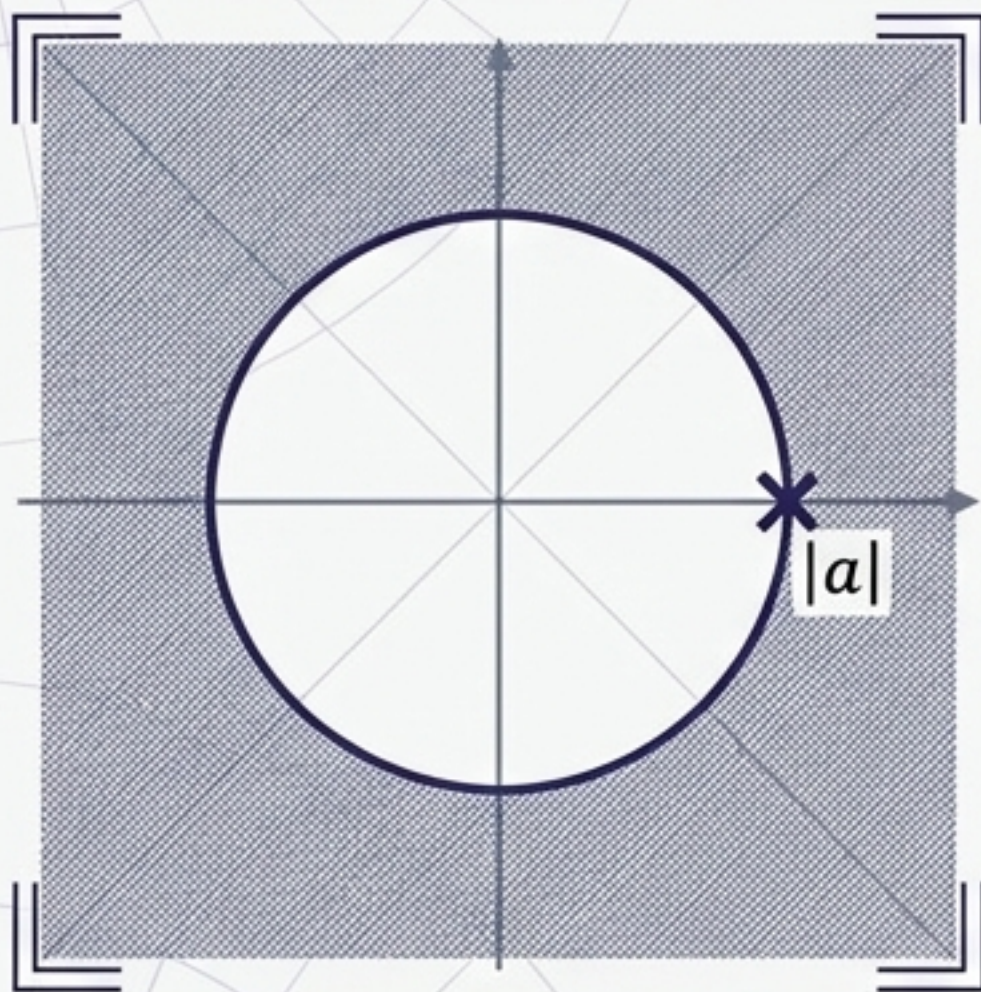
Represents a left-sided sequence:

$$x[n] = -a^n u[-n-1]$$

The algebraic expression is identical; the ROC distinguishes the sequences. Poles are never inside the ROC.

ROC and Causality

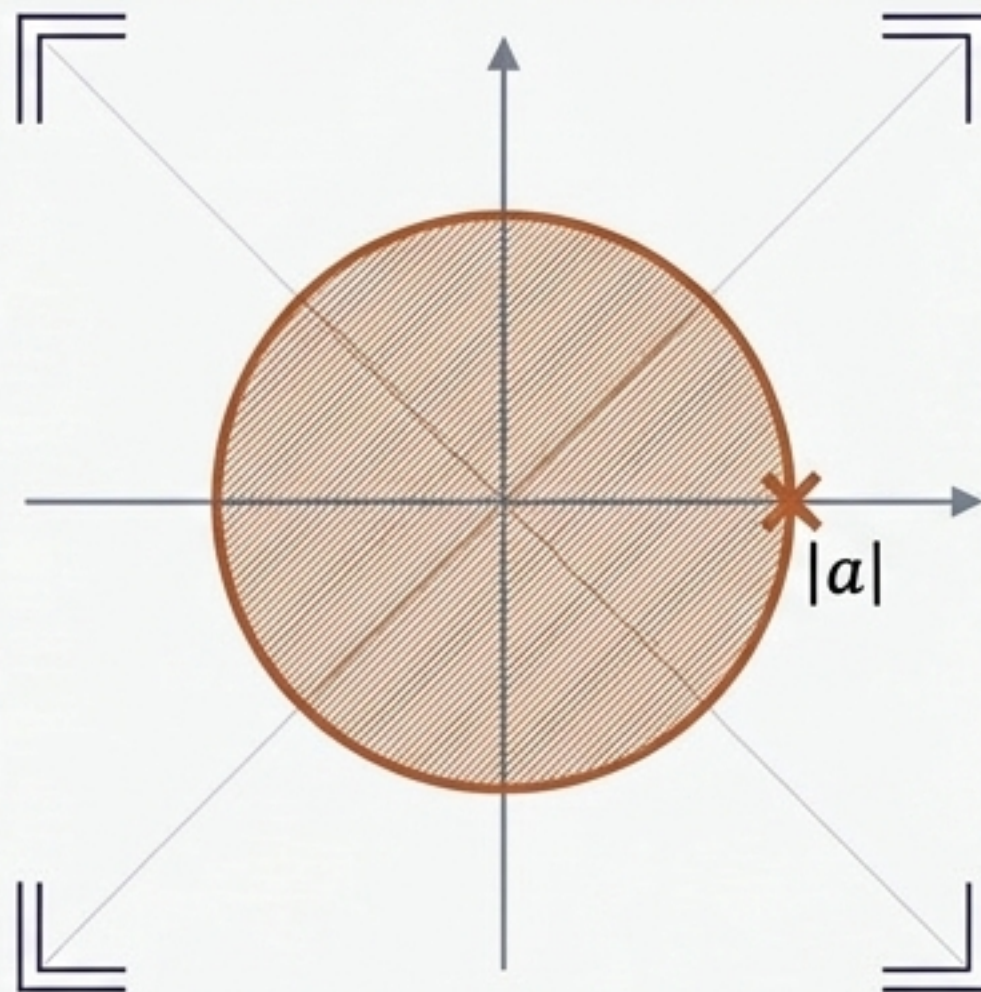
Right-Sided



Condition: $|z| > \max_k |p_k|$
(Outside the outermost pole)

Sequence: Extends infinitely to the right.

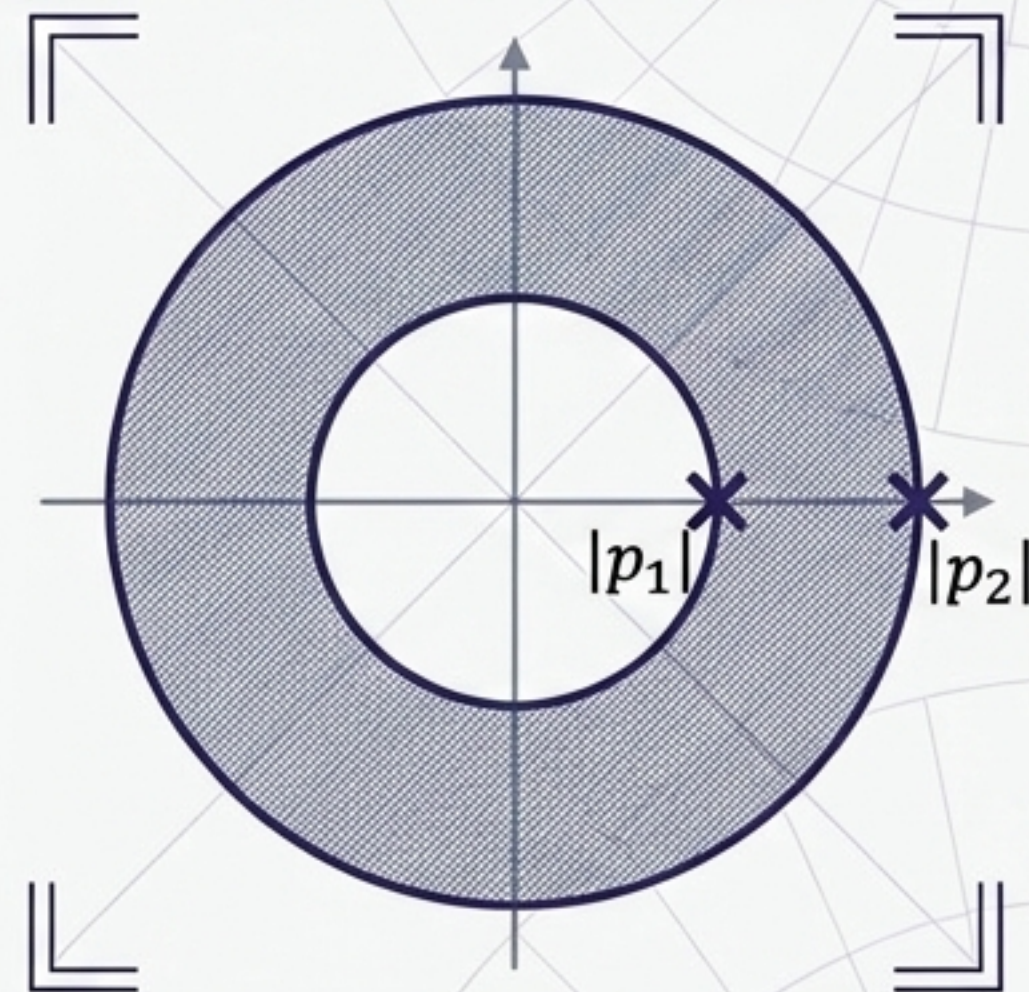
Left-Sided



Condition: $|z| < \min_k |p_k|$
(Inside the innermost pole)

Sequence: Extends infinitely to the left.

Two-Sided



Condition: $|p_1| < |z| < |p_2|$
(Annular region between poles)

Sequence: Extends infinitely in both directions.

$$H(z) = K \frac{\prod_k (z - z_k)}{\prod_m (z - p_m)} = \frac{N(z)}{D(z)}$$

Zeros (O)



Values satisfying $N(z) = 0 \rightarrow H(z) = 0$

Poles (X)



Values satisfying $D(z) = 0 \rightarrow H(z)$ becomes unbounded

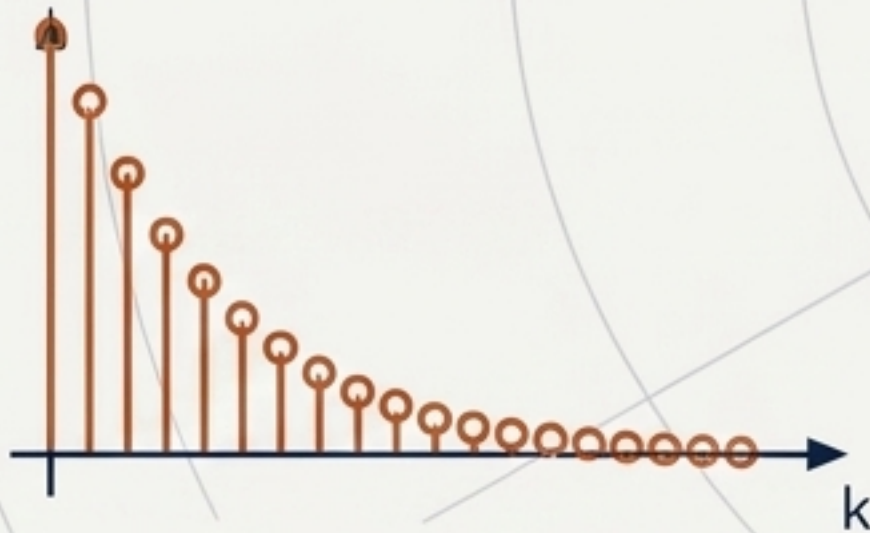
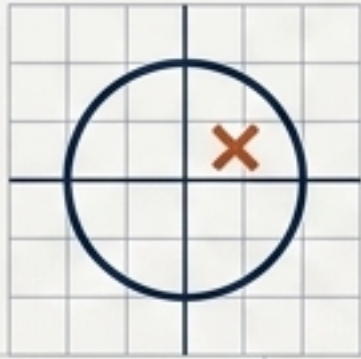
Pole-Zero Topography



These geometric coordinates completely dictate the behavior of the discrete-time system.

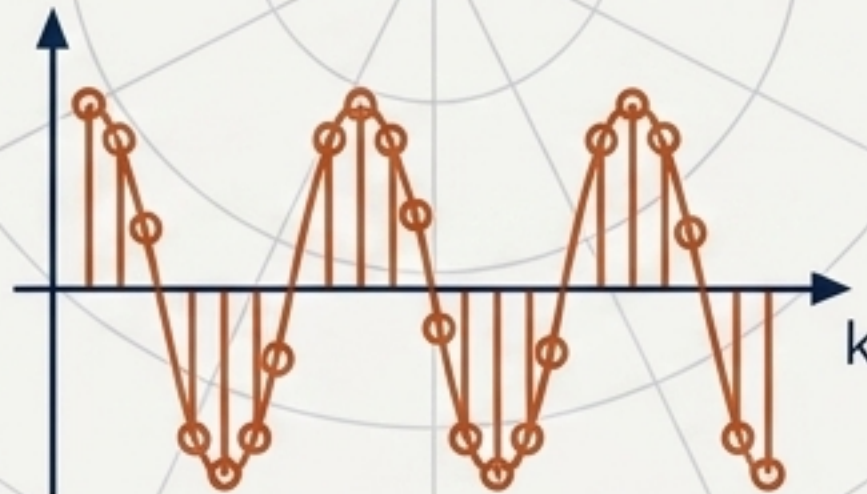
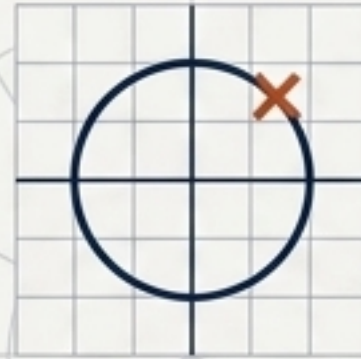
Pole Locations and System Stability Spectrum

Inside Unit Circle
($|p| < 1$)



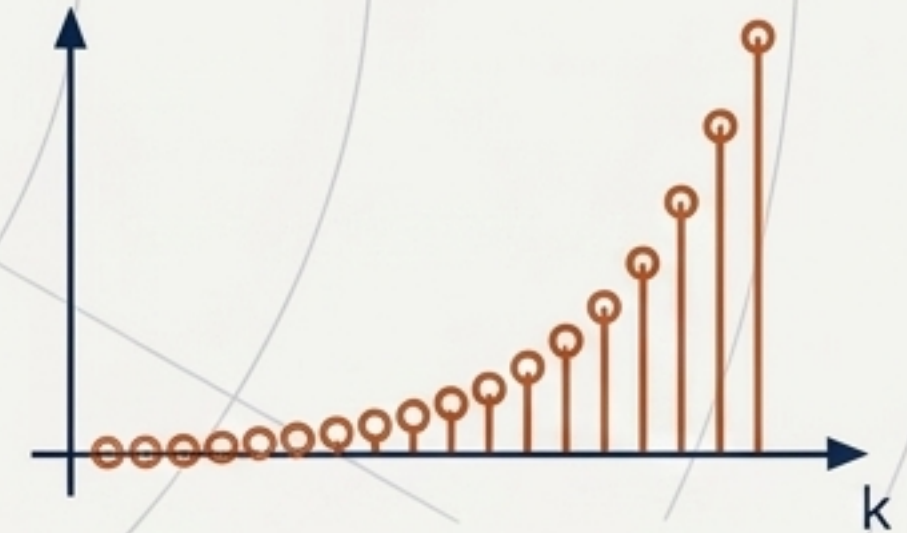
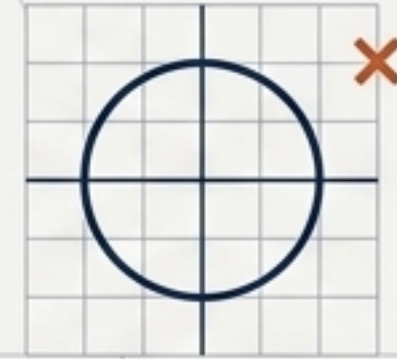
Decay $\rightarrow$ Stable

On Boundary
($|p| = 1$)



Non-decaying $\rightarrow$
Marginally Stable

Outside Unit Circle
($|p| > 1$)

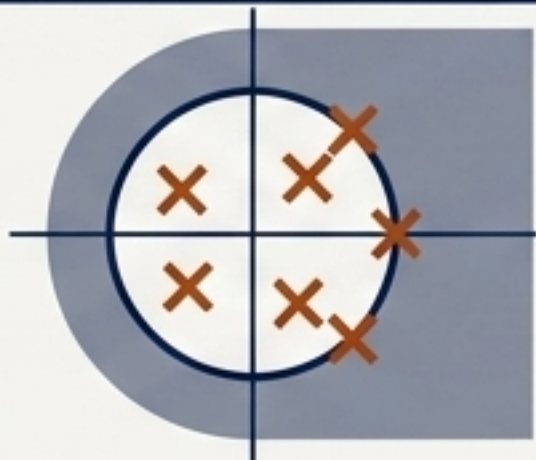


Growth $\rightarrow$ Unstable

A discrete-time LTI system is BIBO stable if and only if:
The Unit Circle ($|z|=1$) is contained in the ROC



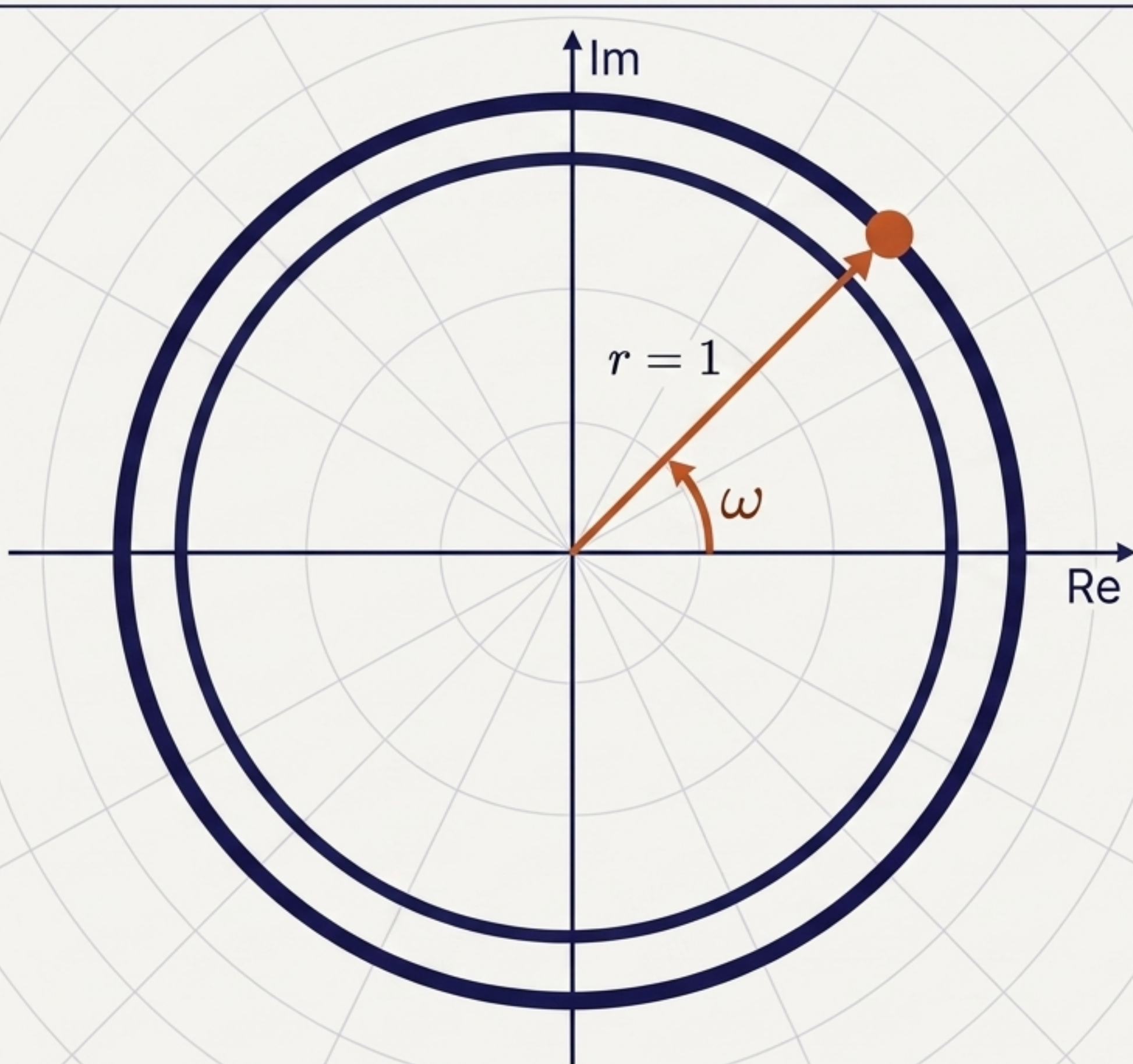
The Rational Causal System Synthesis



For a system that is both **Causal** (ROC is outside outermost pole) and **Stable** (Unit circle in ROC):

All poles p_k must satisfy $|p_k| < 1$

$$|p_k| < 1$$



Transition to the Frequency Domain

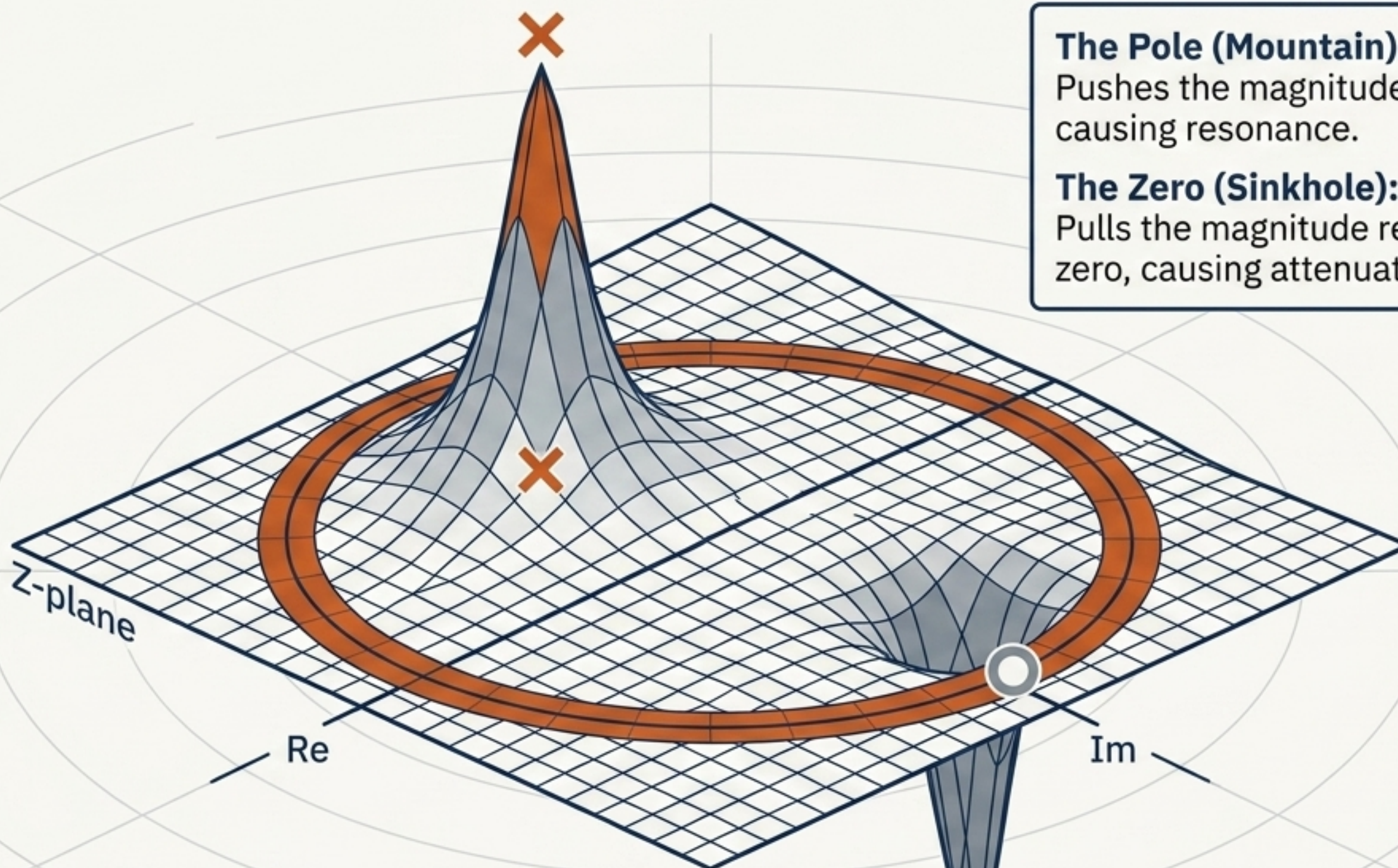
The unit circle is defined by $z = e^{j\omega}$.

If the unit circle belongs to the ROC, the Z-transform collapses into the Frequency Response (DTFT):

$$H(e^{j\omega}) = H(z)|_{z=e^{j\omega}}$$

The Frequency Response consists of magnitude $|H(e^{j\omega})|$ and phase $\angle H(e^{j\omega})$.

The Z-plane provides a physical track to evaluate all frequencies from DC ($\omega = 0$) to Nyquist ($\omega = \pi$).

**The Pole (Mountain):**

Pushes the magnitude response up, causing resonance.

The Zero (Sinkhole):

Pulls the magnitude response to exactly zero, causing attenuation.

Difference Filter

$$y[n] = x[n] - x[n - 1]$$

$$H(z) = \frac{z - 1}{z}$$

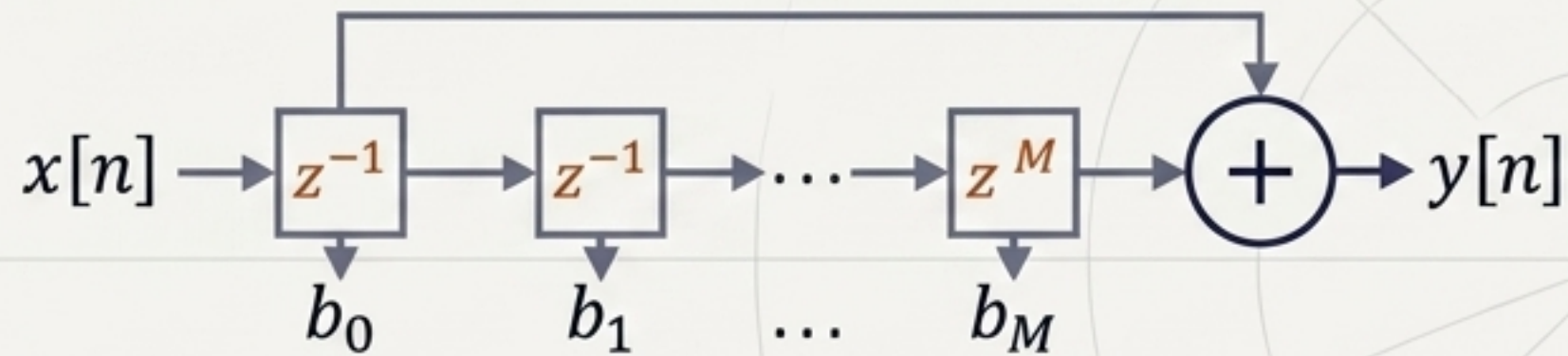
Zero at $z=1$ (which corresponds to DC, $\omega=0$).

This sinkhole exactly at the right edge of the unit circle completely suppresses DC components.

Editorial Diagnostic Blueprint

Digital Filter Architecture Comparison: FIR vs. IIR

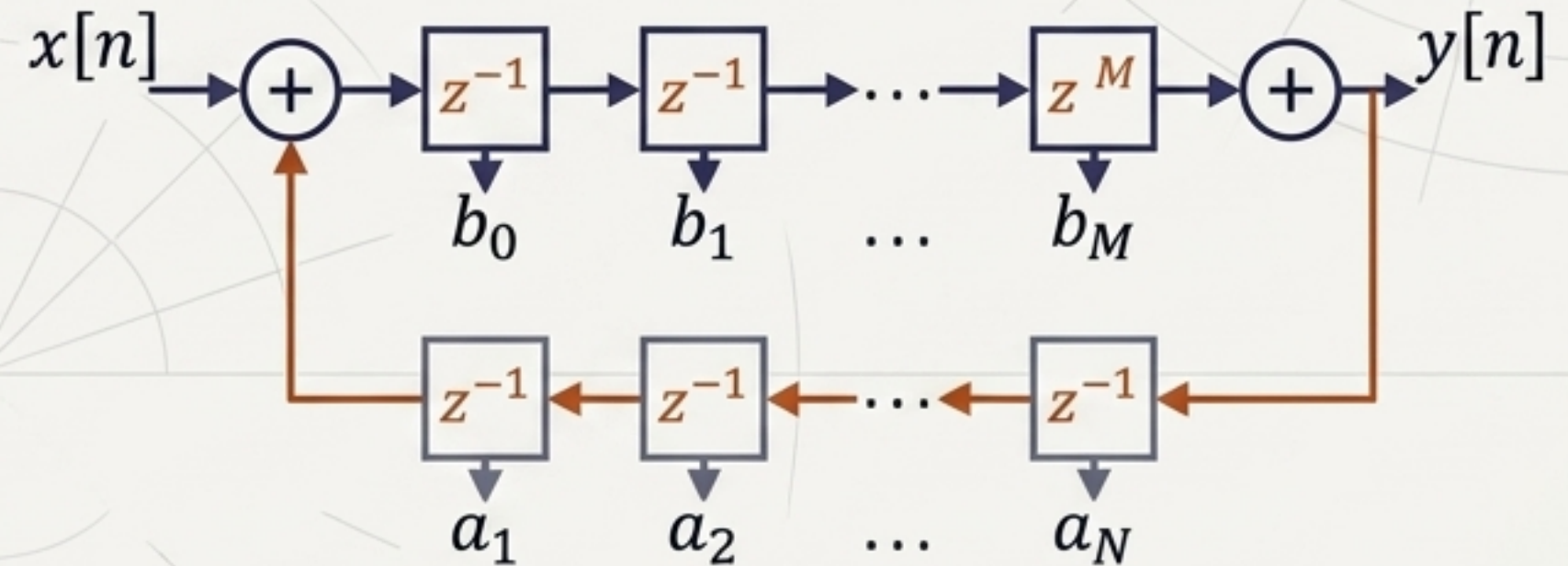
FIR (Finite Impulse Response)



$$H(z) = b_0 + b_1 z^{-1} + \dots + b_M z^{-M}$$

- Impulse response ends after a finite duration.
- Inherently stable (poles only exist at origin $z=0$).

IIR (Infinite Impulse Response)



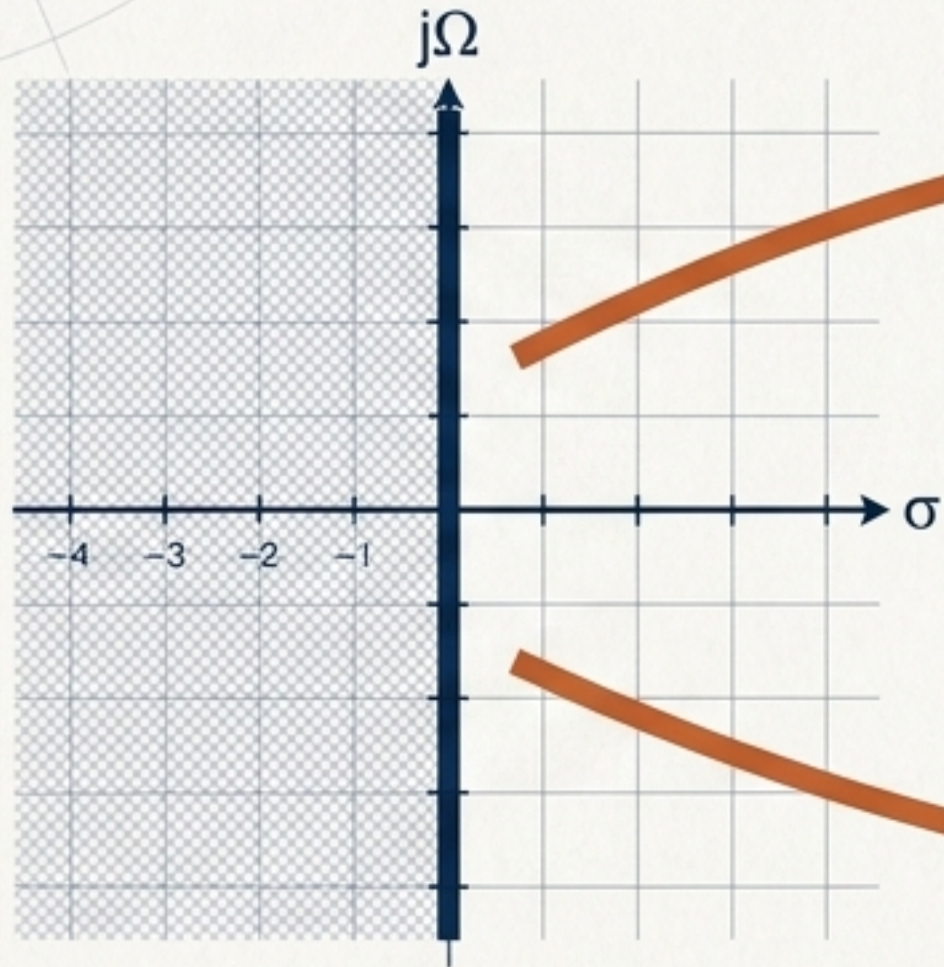
$$H(z) = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

- Potentially infinite response due to feedback architecture.
- Requires rigorous stability checking (poles must be inside unit circle).

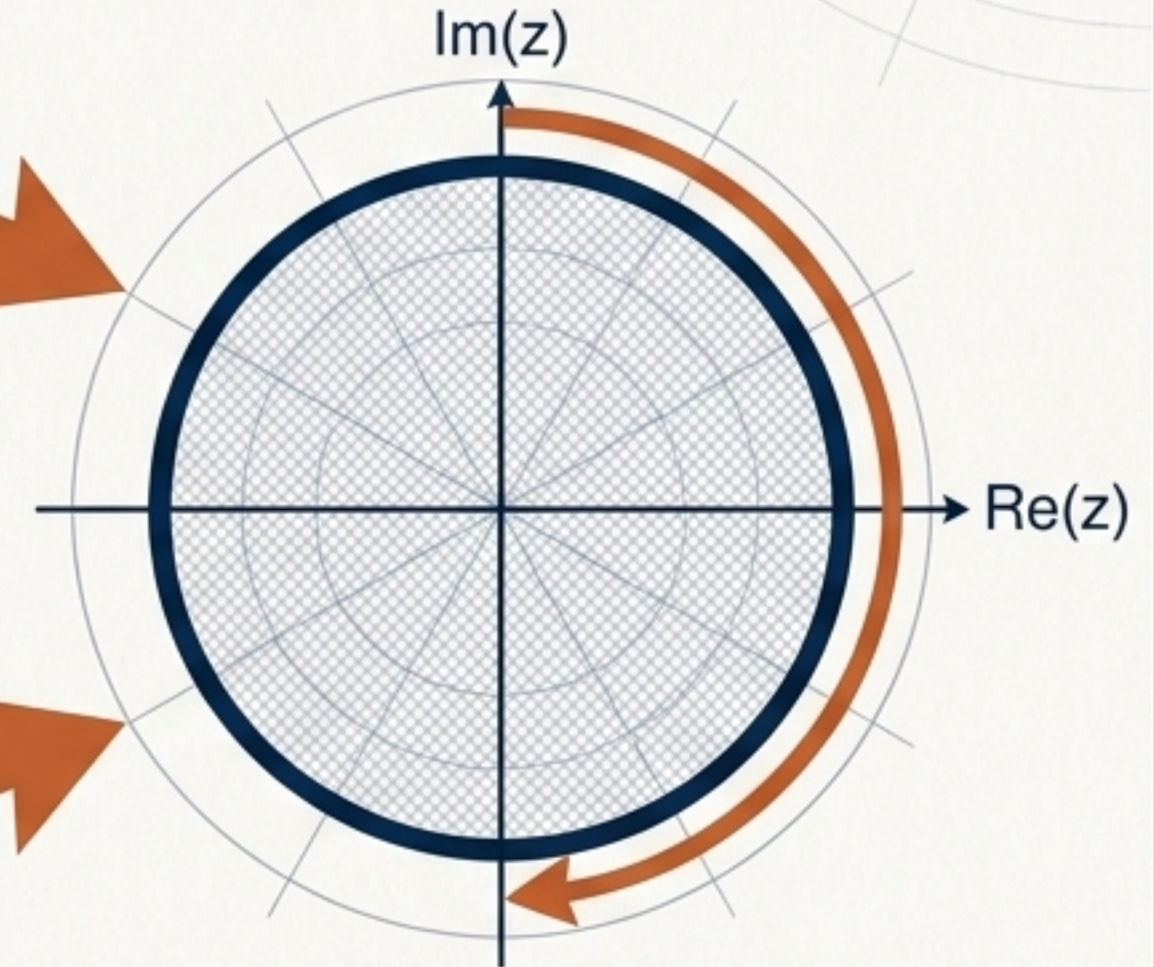
The Laplace-to-Z Portal

$$z = e^{sT}$$

The S-Plane (Continuous)



The Z-Plane (Discrete)



$$\sigma < 0 \iff |z| < 1$$

Continuous-time stability ($\Re(s) < 0$) maps perfectly to discrete-time causal stability ($|z| < 1$).

Reversing the Transformation (Inverse Z-Transform)

Method 1: Known Transform Pairs

Sequence $x[n]$	Transform $X(z)$
$\delta[n - n_0]$	$\leftrightarrow z^{-n_0}$
$u[n]$	$\leftrightarrow 1/(1 - z^{-1})$

Directly match terms to a standard table of Z-transform pairs.

Method 2: Power-Series Expansion

$$X(z) = 1 + az^{-1} + a^2z^{-2} + a^3z^{-3} + \dots$$

Divide out the rational function to reveal the sequence point-by-point.

Method 3: Partial Fractions

$$\frac{1}{(1 - 0.5z^{-1})(1 - 0.2z^{-1})} \rightarrow \frac{A}{1 - 0.5z^{-1}} + \frac{B}{1 - 0.2z^{-1}}$$

Decompose complex rational functions into standard, recognizable terms.

Method 4: Contour Integral

$$x[n] = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz$$

The formal mathematical definition, integrating over a closed path within the ROC.

The Complete Z-Transform Mental Model

